



AXOLOTL · AMBYSTOMA MEXICANUM

Cold water is the whole job

A complete, fact-checked printable manual: cold-water tank setup, cooling, cycling, water chemistry, diet, health triage, and the routines that keep an axolotl thriving for its full lifespan.

INSIDE

- ✓ Cooling a tank without a chiller, and when you need one
- ✓ Fishless and animal-in cycling, step by step
- ✓ Health triage: gills, fungus, impaction and floating
- ✓ Owner logs, pet-sitter sheet and a heat wave plan

41 pages

Intermediate

Print or tablet

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Every page is written to be used, not just read once.

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START HERE

How to use this package

Four ways to get value from it, depending on where you are right now.

If you haven't set up yet

Read pages 4 through 14 in order before buying anything, and check page 6 first if you are in California, New Jersey, Maine or Washington DC. Cooling is the decision that shapes the whole budget, and it is far cheaper to plan for than to retrofit in July. Expect to spend several weeks cycling the tank before an axolotl goes in it. Then use the Budget and Shopping List on page 31 to buy once.

If your axolotl is already set up

Jump to the Setup Checklist and Targets on page 29 to audit what you have, and the Health Red Flags section starting on page 23 if anything seems off. Check the temperature and test the water before you change anything else. Most axolotl problems that look like disease are a thermometer reading or a water reading, and you cannot rule either out by looking at the animal.

If the water is already too warm

Go straight to page 8. Warmth is the single most common way a pet axolotl is harmed, and the fixes that work in the next hour are not the same as the ones that work all summer. Do not drop the temperature all at once, and do not add a heater at any point for any reason.

Print these first

- | | |
|--|---|
| ☐ Page 29: Setup Checklist & Targets | ☐ Page 36: Pet-Sitter Sheet |
| ☐ Page 30: Emergency & Quick Targets Card | ☐ Page 37: Owner Log |
| ☐ Page 33: Symptom Quick Reference | ☐ Page 38: Equipment & Maintenance Log |
| ☐ Page 34: Daily, Weekly & Seasonal Routine | ☐ Page 39: Enrichment Checklist & Log |

WHY THIS PACKAGE EXISTS

Axolotl advice online contradicts itself most on the two things that matter most: how cold the water has to be, and what is safe to put on the tank floor. This package gives numbers you can test against, cross-checked against veterinary and research-colony sources, listed on page 41. Every temperature and dimension is given in both imperial and metric.

NOT A SUBSTITUTE FOR

An amphibian or aquatic veterinarian. This is a husbandry reference for day-to-day care. Anything urgent or health-related is always a professional call, not a page in this book. Vets who see axolotls are rarer than small-animal vets, so find one before you need one and write the number on page 30.

SECTION 01

Quick profile

Everything at a glance before you read the detail.

Scientific name	<i>Ambystoma mexicanum</i> , a mole salamander and a close relative of the tiger salamanders
Adult size	Most reach 9 to 10 in (23 to 25 cm); some individuals reach about 12 in (30 cm)
Captive lifespan	10 to 15 years with consistent care. Wild lifespans are shorter, and are not the number to plan around
Sexual maturity	About 1 year
Water temperature	60 to 68°F (16 to 20°C), with the cooler half of that band preferred. Never above 72°F (22°C)
Tank minimum	20 gal long (78 L) for one adult, 40 gal breeder for two. Juveniles can start in 10 gal (38 L)
Water chemistry	Ammonia 0 ppm, nitrite 0 ppm, nitrate under 10 ppm, pH 7.4 to 7.6, moderately hard water
Diet type	Carnivore. Earthworms and nightcrawlers are the nutritionally complete staple
Feeding	Juveniles daily, adults every 2 to 3 days
Housing	Species-only, no fish. Juveniles alone, adults in pairs only with volume and hides
Handling	Medical necessity only. This is a look-at pet, not a hold-it pet
Wild status	Critically Endangered (IUCN). Effectively every pet axolotl is captive-bred

THE ONE-LINE VERSION

An axolotl is a cold-water aquarium animal that gulps whatever fits in its mouth. Keep the water cold, keep the floor safe to swallow, and most of the rest of this package is maintenance.

What "never grows up" actually means

An axolotl is paedomorphic: it reaches full adult size and breeds normally while keeping the feathery external gills, tail fin and fully aquatic life of a larva. That is not the same as staying a baby, and it is not quite permanent either. The tissues still answer to thyroid hormone, and researchers can push an axolotl through the land-going metamorphosis a tiger salamander does on its own. What is switched off is the hormonal trigger, not the ability. In a home tank that switch essentially never flips, which is why nothing in a correct axolotl setup needs a land area or a basking spot.

SECTION 01

Cost, commitment & fun facts

The animal is the cheap part. Cooling is what sets the budget.

THE AXOLOTL

\$30 to \$100 for a standard morph. Rarer colors such as mosaic or GFP run higher.

SETUP, NO CHILLER

\$200 to \$500, and only honest if your room genuinely holds under 68°F (20°C) year round.

SETUP WITH A CHILLER

\$450 to \$900. The chiller alone is \$150 to \$400 and is the line that separates the two ends.

MONTHLY

\$15 to \$45 for food, conditioner, filter media and the chiller's electricity.

What you are actually committing to

COMMITMENT	WHAT IT MEANS IN PRACTICE
10 to 15 years	Longer than most people expect from a small aquatic animal. Plan for house moves, college, and summers away.
A cold room, or a chiller	The cost that varies most by climate and by season. A warm apartment in August is the scenario to budget for, not the mild week you buy the tank in.
Weekly testing and water changes	Around 20 minutes a week once the routine is set, plus five minutes daily to check temperature and appetite.
A vet you have already found	Amphibian-experienced vets are uncommon. Finding one after something goes wrong wastes the days that matter most.

BUDGET FOR THE BAD MONTH TOO

Surgery for a swallowed gravel impaction, or a course of treatment for a severe bacterial or fungal infection, costs well beyond routine care. Neither is likely if you follow this package, and both are real enough to keep a small fund for. Ask your vet for a specific estimate rather than guessing.

Fun facts worth knowing

- Axolotls regenerate limbs, and also spinal cord, heart and brain tissue, without scarring. A gill filament nipped by a tank mate usually regrows.
- They have no eyelids. Dim, indirect light is not a preference, it is the condition they are built for.
- They hunt by scent and water movement rather than sight, which is why a worm offered by tongs works better than food dropped anywhere in the tank.
- The species is Critically Endangered in its only native home, the Xochimilco canal system in Mexico City, while millions live in tanks and research colonies worldwide.
- A glass or mesh lid is not decoration. An axolotl can get itself out of an open tank.

SECTION 01

Where axolotls are legal

A short list of real restrictions, and one myth that outlived its own repeal.

JURISDICTION	STATUS	BASIS
California	Banned	Cal. Code Regs. tit. 14, section 671 lists genus <i>Ambystoma</i> as a detrimental restricted species. No pet permits are issued.
New Jersey	Effectively banned	State environmental protection guidance lists axolotls among species that may not be kept as pets or sold. Permits are not issued for hobby purposes.
Maine	Restricted, not permitted as pets	12 M.R.S. section 12152. Axolotls are not on the unrestricted list and possession permits are not typically issued for pet ownership.
Washington, DC	Reported banned	District prohibited-animals rules.
New Mexico	Import-restricted	Possession is allowed in some cases, but importing from other states is restricted.

VIRGINIA IS LEGAL, AND HAS BEEN SINCE 2021

Virginia is still listed as a ban state across a surprising number of hobbyist sites and forum posts. Since August 1, 2021 it has been legal to import, export, sell and possess axolotls there with no permit, following an amendment that carved axolotls out of the state's general salamander-sale prohibition. If you have seen Virginia on a ban list, that list predates the change.

Why the bans exist

None of the restrictions above are about protecting axolotls. They protect native salamanders. Axolotls sit in genus *Ambystoma* alongside North America's mole and tiger salamanders, including California's federally threatened California tiger salamander, so regulators weigh hybridization, an escaped population establishing itself, and disease. California's "detrimental species" label makes that explicit.

The 2025 federal rule, and what it does not do

A federal rule effective January 10, 2025 added 16 more salamander genera to the injurious wildlife list under the Lacey Act, bringing the total to 36, all tied to the salamander chytrid fungus *Batrachochytrium salamandrivorans* (Bsal). Genus *Ambystoma* is among the newly added genera.

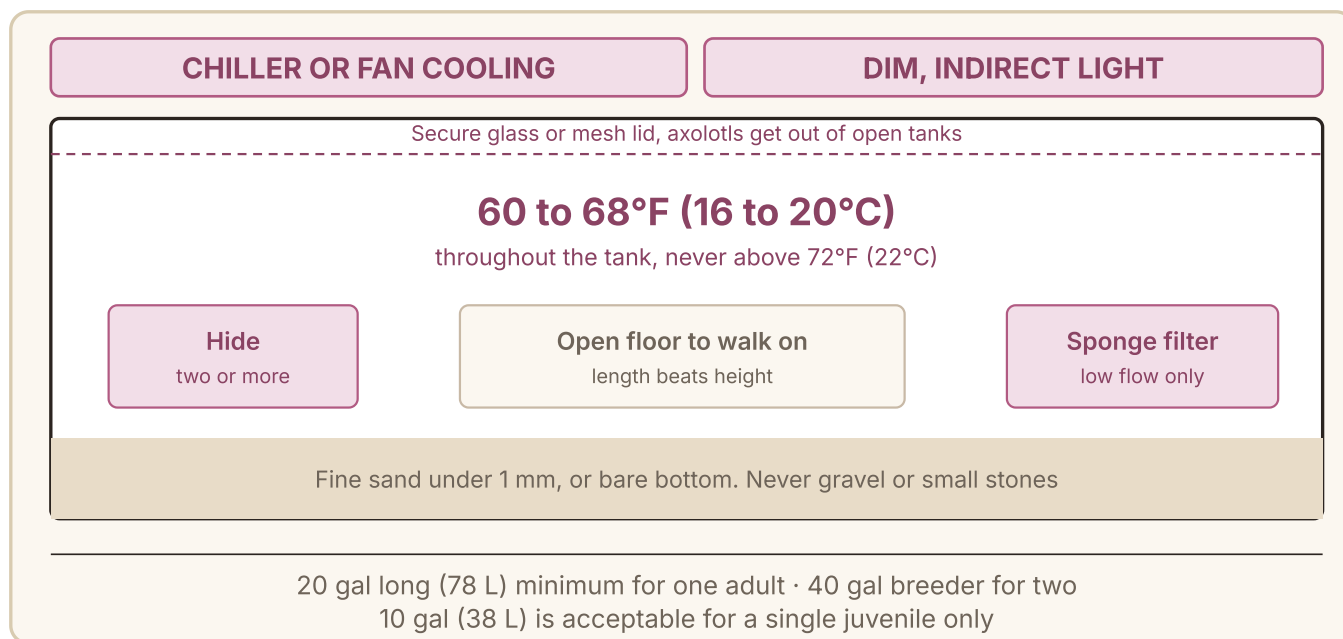
What that means for an owner: the listing restricts importing axolotls into the US and shipping them to or from Hawaii, DC, Puerto Rico and US territories. It does not restrict owning, breeding or selling one within the 49 continental states, a distinction the US Fish and Wildlife Service states directly in its own FAQ on the rule.

CHECK BEFORE YOU BUY, NOT AFTER

Laws move, and a shipper's willingness to send one is not a legal opinion. Confirm your state and city rules before you buy a tank, and never release an axolotl or tank water into any waterway. Bsal is why that rule exists.

Tank size & the cold-water setup

A 20 gallon long (78 L) tank is the minimum for one adult, and a 40 gallon breeder is the better answer for two. Floor space matters far more than height: an axolotl is a bottom-dweller that walks more than it swims and only surfaces for a gulp of air. Water volume is not just space, it is stability. The larger the volume, the slower temperature and water chemistry drift, and drift is what actually harms this animal.



NEVER

Put a heater in an axolotl tank. For almost every other aquarium animal warmth is the fix. Here it is the hazard, and the equivalent piece of equipment is a chiller or a fan, covered on page 8.

What the diagram does not show

- Put the tank somewhere quiet and out of direct sun. Constant movement past the glass is a stressor, and a sunlit tank is a heat problem you cannot fan your way out of.
- Cycle the tank fully before an axolotl goes in it. Ammonia and nitrite both have to hold at 0 ppm. See page 11.
- Stand-height matters for you, not the animal. Water changes are a weekly job, so put the tank where you can reach the bottom with a siphon.
- Leave a gap of an inch or two (a few cm) of air below the rim. Axolotls take occasional gulps of air at the surface.

Temperature: the numbers that matter

The single most important number in this package, and the one most often missed.

READING	WHAT IT MEANS
60 to 68°F (16 to 20°C)	The keeping range. Veterinary guidance prefers the cooler half, 60 to 64°F (16 to 18°C), and a research colony targets 60 to 65°F (15 to 18°C).
Above 72°F (22°C)	The ceiling. Sustained warmth past this point suppresses the immune system and stress becomes chronic rather than occasional.
Above 75°F (24°C)	High probability of illness. Expect appetite loss, uncontrolled floating, and bacterial or fungal infection. Prolonged exposure here kills.
Below about 50°F (10°C)	Cold slows an axolotl down and cuts its appetite. Not an emergency the way heat is, and vets use cooler water as treatment, but a permanently cold tank is not the goal either.

Why warm water is so dangerous here

Three things happen at once. The immune system falters, so bacteria and fungi a healthy axolotl shrugs off get a foothold. Warm water holds less dissolved oxygen, and this animal breathes through exposed external gills. And those same bacteria grow faster in the warmth. Heat stress rarely arrives as a single symptom, which is why cooling the water is the first treatment for problems that look like something else.

MEASURE WHERE THE ANIMAL IS

Put the thermometer in the water at the bottom of the tank, not on the outside glass, and read it at the hottest part of the day. A tank that holds 65°F (18°C) at 8am and 74°F (23°C) at 5pm is not a 65°F tank.

Which method your situation needs

YOUR ROOM	WHAT TO PLAN FOR
Under 68°F (20°C) all year, such as a basement	A thermometer and a fan on standby is enough. Check it through a full summer first.
Cool most of the year, a few hot weeks	Fan, frozen bottles and air conditioning in that room. Have all three ready in spring, not on the day.
In the 70s°F (low 20s°C) for weeks at a time	A chiller. Nothing else holds a number that long, and a tank that spends every August above 72°F (22°C) is not one an axolotl survives for a decade.
Hot climate, no air conditioning	A chiller as core equipment, in the budget from the start, with a plan for the outlet, the noise and the exhaust heat.

Cooling methods & the summer plan

Six ways to take heat out of a tank, cheapest first.

Cooling methods, cheapest first

METHOD	BUYS YOU	NOTES
Move the tank	Varies	Out of sun, off an exterior wall, onto a lower floor or into a basement. Free, and often the largest single drop available.
Cut heat sources	1 to 2°F	Dimmer light, shorter photoperiod, and check whether the filter or pump motor itself is warming the water.
Fan across the surface	2 to 5°F	Evaporative cooling. Aim a small fan over an open or partly open lid. Costs water, so top up with dechlorinated water and watch that the level does not drop.
Insulate the tank sides	A few °F	Helps most in a room that swings between day and night. Slows the gain rather than removing it.
Room air conditioning	Whatever you set	Reliable, and cools you too. Expensive to run all summer in a hot climate.
Aquarium chiller	Sets a target	The only method that holds a number regardless of the weather. \$150 to \$400, plus running cost. In a warm climate it is core equipment, not an upgrade.

FROZEN BOTTLES ARE A STOPGAP, NOT A PLAN

A sealed bottle of frozen dechlorinated water floated in the tank does drop the temperature, and it is worth knowing for a heat wave or a blackout. It is a poor routine method: the swing as it melts and gets replaced is itself a stressor, and it demands someone at home to keep swapping bottles. Never tip loose ice cubes into the tank. Aim for a change of no more than 1 to 2°F (about 1°C) an hour in either direction.

Buying and running a chiller

- Size up rather than down. A unit working at its limit runs constantly and leaves you no margin on the hottest day of the year.
- Water has to be pumped through it, so check your pump's flow rate falls inside the chiller's rated range.
- It exhausts the heat it removes into the room. Leave clearance, and keep it out of a closed cabinet.
- They are not silent, which matters if the tank is in a bedroom.
- Set the target mid-range, not at the edge, so normal cycling never takes the water out of range.

TEST THE PLAN BEFORE YOU NEED IT

Run your method against the worst conditions you expect while the tank is still empty, or at least before summer. A setup only ever tested in April is untested, and the failure shows up as a sick animal rather than an error message.

Filtration & flow

Enough filtration to process a messy carnivore's waste, delivered gently enough that it does not push the animal around.

Axolotls want slow, near-still water. They come from a canal system, not a stream, and a filter sized by flow alone will stress one badly. At the same time this is a large carnivore eating whole worms, so the waste load is real and the filter has to keep up. The way out is not a weaker filter. It is an adequately sized filter whose output has been broken up before it reaches the water.

The three jobs a filter does

TYPE	WHAT IT DOES, AND WHY AN AXOLOTL TANK NEEDS IT
Mechanical	Traps uneaten worm, shed slime and waste before it breaks down. Strong mechanical filtration is worth having here, because leftover food is the main way water quality collapses in this tank.
Biological	The bacteria colony that turns ammonia into nitrite and nitrite into nitrate. This is the part you protect at all costs. It lives in the media, not the water.
Chemical	Carbon and similar media. Useful for clearing medication after treatment, not needed day to day. Remove carbon while dosing anything, since it strips the medication out.

Choosing and taming the filter

Good choices

- A large sponge filter run off an air pump. Gentle by design, cheap, and it doubles as an air source in a power cut.
- A canister rated for the tank or a size up, with the return sent through a spray bar aimed along the back glass or at the surface.
- A hang-on-back unit with the outflow baffled, for example against a piece of filter foam or a bottle-cut deflector.

Signs the flow is too strong

- Gills swept back flat and held there rather than fanning out.
- The axolotl parked in one corner permanently, or repeatedly pushed off the glass.
- Sand blown into drifts, or the animal bracing against the current to stay put.
- Food carried away from the animal before it can find it.

RINSE MEDIA IN TANK WATER, NEVER TAP WATER

Chlorine and chloramine kill the bacteria colony you spent weeks growing. When mechanical media clogs, swish it in a bucket of water siphoned out of the tank and put it straight back. Never replace all the media at once, and never run a brand new filter alongside nothing else and call the tank cycled.

FILTRATION DOES NOT REPLACE WATER CHANGES

A filter converts ammonia and nitrite into nitrate. It does not remove the nitrate, and does nothing about what else builds up between changes. That is what page 13 is for.

Cycling, with or without an axolotl

The weeks of work that happen before the animal, and the recovery plan if it did not.

Cycling means growing the bacteria that convert ammonia to nitrite, and nitrite to nitrate, inside your filter media. Until that colony exists the tank cannot process waste, and ammonia and nitrite accumulate to levels that burn skin and gills. There is no shortcut and no way to tell by looking: the only evidence is a test kit reading zero on both while the tank is being fed ammonia. Expect four to six weeks.

Fishless cycle, the right way to do it

STEP	WHAT TO DO
1	Set the tank up complete: filter running, substrate in, hides placed, water dechlorinated, cooling running and holding temperature.
2	Add an ammonia source and bring the tank to a measurable reading. Pure ammonia dosed to 2 to 4 ppm is the controllable way. Seed the filter with media, a sponge or gravel from an established, healthy tank if you can get it, which can cut weeks off the process.
3	Test every day or two. Ammonia rises, then falls as nitrite appears. Nitrite then rises and falls as nitrate appears. Top the ammonia back up whenever it reads zero, so the colony keeps being fed.
4	The completion test: dose ammonia to about 2 ppm, wait 24 hours, and test again. Ammonia 0 and nitrite 0 with nitrate present means the tank is cycled.
5	Do a large water change to bring nitrate under 10 ppm, confirm the temperature is holding in range, and only then bring the animal home.

IF THE CYCLE STALLS

Nitrite that will not clear is the usual sticking point. Check that the temperature has not dropped so low that bacterial growth crawls, that you have not rinsed media in tap water, that you are not overdosing ammonia past about 5 ppm, and that pH has not crashed. Patience beats additives here. A cycle that looks stuck is often three days from finishing.

If your axolotl is already in an uncycled tank

This is a common situation and it is fixable. Do not tear the tank down, do not replace the filter, and do not start dosing treatments. The animal itself is now the ammonia source, so the cycle will complete on its own. Your job is to keep the readings survivable while it does.

- Test ammonia and nitrite every day, at the same time of day.
- Change 25 to 50 percent of the water with dechlorinated, temperature-matched water whenever either reads above 0.25 ppm.
- Feed less and remove every scrap of uneaten food. Less input, less ammonia.
- Use a conditioner that temporarily detoxifies ammonia and nitrite between changes. It buys time, it does not cycle the tank.
- Consider tubbing the animal in daily-changed water while the tank finishes, as on page 28. For a badly overloaded tank this is often kinder than the water changes.

Water targets & testing

Six numbers. Two of them are the vital signs.

PARAMETER	TARGET	ACT AT	WHY IT MATTERS
Temperature	60 to 68°F (16 to 20°C)	Above 72°F (22°C)	Immune suppression, then infection. See page 8.
Ammonia	0 ppm	Any reading at all	Burns skin and gills, damages organs. There is no safe low level.
Nitrite	0 ppm	Any reading at all	Blocks the blood's ability to carry oxygen.
Nitrate	Under 10 ppm	Above 20 ppm	Chronic exposure is linked to bulging eyes and general poor condition. Water changes are the only real fix.
pH	7.4 to 7.6	Outside 6.5 to 8.0	Stability matters more than the exact figure. At high pH, ammonia is more toxic than the same reading at neutral.
Hardness	Moderately hard	Very soft or distilled	Axolotls need mineral content to maintain skin integrity. Soft water also lets pH swing.

A testing schedule that is actually sustainable

WHEN	TEST
Daily	Temperature. Ten seconds, and it is the reading most likely to change without warning.
Weekly	Ammonia, nitrite, nitrate and pH, on the same day each week, before the water change rather than after.
Daily, temporarily	Ammonia and nitrite while a tank is cycling, after adding a second animal, after a filter change, or any time something looks wrong.
Every six months	Check the test kit's expiry date. Old reagents read low, which is the worst possible failure mode.

USE A LIQUID TEST KIT, NOT STRIPS

Paper strips are convenient and consistently imprecise, particularly at the low end where it matters most here. A liquid drop-based master kit covers ammonia, nitrite, nitrate and pH for hundreds of tests and costs about the same as two months of strips.

TAP WATER IS NOT JUST WATER

Municipal supplies carry chlorine or chloramine, and chloramine does not gas off if you leave a bucket standing. Every drop of new water gets a dechlorinator, every time, including top-ups for evaporation. If your supply is very soft or you use RO water, it needs remineralizing before it goes near an axolotl.

Water changes & early warnings

The one job that fixes more problems than every product on the shelf combined.

The routine

SITUATION	CHANGE
Established, cycled tank	20 to 30 percent once a week. Veterinary guidance puts the figure at about 30 percent weekly, and splitting it into two smaller changes is gentler if nitrate is climbing.
Nitrate still above 10 ppm after a change	Increase the volume or the frequency, not the number of additives. Check that you are not overfeeding first.
Cycling with an animal in the tank	25 to 50 percent whenever ammonia or nitrite reads above 0.25 ppm, daily if that is what it takes.
After medicating	As directed by the vet, plus running carbon afterward to clear the residue.

How to do one without stressing the animal

1. Prepare the new water first: dechlorinate it and match it to the tank's temperature. A bucket of cold tap water poured straight in is a shock.
2. Siphon from the bottom, lifting settled waste and leftover food off the substrate rather than taking clean water off the top.
3. Leave the axolotl in the tank. Netting it out stresses it more than the change does, and nets snag gills and toes.
4. Refill slowly, pouring onto a plate, a hide or your hand so the stream does not blast the substrate or the animal.
5. Log the date, the percentage and the readings on page 37.

Early warnings, in the order you will notice them

WHAT YOU SEE	WHAT IT USUALLY MEANS
Gills curling forward toward the face	Stress, most often heat or water quality. The earliest visible sign there is, and worth checking daily.
Gill filaments shortening or looking sparse	Longer-running poor water quality, or too much flow. Gills regrow once conditions are corrected.
Appetite dropping off	Temperature or water chemistry, before it is anything else.
Sitting at the surface, or floating and unable to sink	Gas, impaction or heat. See pages 26 and 24.
Red or inflamed patches on skin or gills	Ammonia or nitrite burn. Test immediately and change water. See page 27.

NEVER DO A FULL TEARDOWN

Emptying a tank, scrubbing everything and refilling it destroys the bacteria colony and puts you back at week one of cycling, usually just as the animal is already unwell. Siphon the substrate and rinse mechanical media in tank water instead.

Substrate, hides & decor

An axolotl swallows whatever fits in its mouth, so the tank floor is a safety decision, not a styling one.

Axolotls feed by gulping: they open their mouth and inhale everything in front of it, food and floor alike. That single fact decides the substrate. Anything small enough to fit in the mouth and too large to pass through the gut is an impaction waiting to happen, and gravel impaction is one of the most common reasons a pet axolotl ends up in surgery.

SUBSTRATE	VERDICT	DETAIL
Bare bottom	SAFEST	Zero impaction risk and the easiest to keep clean. The surface is slick, so some axolotls slip on it, and it shows every scrap of waste, which is arguably a feature. The right answer for any juvenile under about 5 in (13 cm).
Fine sand, under 1 mm	GOOD	Natural to walk and forage on, and fine enough to pass through the gut if swallowed. Not risk-free, so keep it shallow, siphon it every water change, and use it only for adults.
Large smooth stones	SITUATIONAL	Safe only if every stone is genuinely larger than the animal's head. Waste and uneaten food collect between them, so water quality is the trade.
Gravel, pebbles, coarse sand	NEVER	The classic fatal mistake. Swallowed, too big to pass, and the resulting blockage often needs surgery.

Hides and furnishings

Use

- At least two hides per adult, ideally three or four of different shapes, so the animal can choose light level and company.
- Terracotta pots on their side with the drainage hole sealed, wide PVC sections with every edge sanded, and smooth resin or ceramic caves.
- Live or silk plants: java fern and anubias tied to decor rather than buried, and moss balls larger than the animal's head.
- Driftwood and large stones, positioned so they cannot topple or trap a limb.

Avoid

- Anything with a sharp edge, a rough cast seam or a gap a limb can wedge into.
- Plastic plants with stiff, cut edges. Skin this thin tears on things a fish would ignore.
- Small ornaments, marbles, glass beads and anything an axolotl could inhale.
- Bright lighting of any kind. If plants need strong light, the animal needs somewhere shaded to sit out of it.

THE HEAD TEST

Before anything goes in the tank, hold it up and ask one question: could this fit in the animal's mouth, and would it pass through if it did? Larger than the head, or fine enough to pass, are the only two safe answers. Everything in between is the category that causes impactions.

Choosing an axolotl, morphs & sexing

What to look at before you pay, what the colors actually are, and how to tell males from females.

What to check before you buy

Good signs

- Full, feathery gills fanning outward and backward, with long filaments.
- Clear skin with no fuzzy patches, red streaks, sores or missing toes.
- A rounded body roughly as wide as the head, and a straight, unkinked tail.
- An alert response to a passing shadow, and a seller willing to feed one in front of you.
- A cool tank at the shop, and a known hatch date and current diet.

Walk away from

- Warm water, gravel substrate, or fish sharing the tank.
- Curled-forward gills, or gills worn down to stubs.
- White cottony growth anywhere on the body or gills.
- A floating animal, or one lying on its side.
- Mixed larvae with visibly different head shapes. Tiger salamander larvae look similar young, then metamorphose onto land.

Morphs, in plain terms

Morphs are color varieties, not different species or different care. Every one of them needs exactly the same cold water. The common ones are wild type (mottled brown-green with gold flecking), leucistic (pale pink body with dark eyes, the classic pet-shop axolotl), white albino (pale with clear or pink eyes), golden albino (yellow-gold), melanoid (uniformly dark, no gold flecking) and copper (soft tan). Rarer combinations such as mosaic, chimera and the fluorescent GFP line cost more and are cosmetic only. One practical note: a dark morph hides early redness, fungus and gill color change, so a melanoid needs a closer look under decent light.

Sexing

FEATURE	MALE	FEMALE
Cloaca (the vent behind the back legs)	Noticeably swollen and bulging	Flat, or only slightly raised
Body shape	Relatively straight and slim	Rounder and plumper, particularly when carrying eggs
Nails	Often dark or black on the toe tips	Usually pale
Tail	Tends to be longer	Tends to be shorter

None of this is reliable before maturity at about a year, so a juvenile sold as sexed is a guess.

A PAIR WILL BREED, AND THAT IS A BIGGER COMMITMENT THAN IT SOUNDS

A mature male and female housed together will produce clutches of several hundred eggs, repeatedly. The larvae are cannibalistic, need separating and need live food several times a day. If you do not want to raise hundreds of animals or dispose of eggs, keep a single axolotl or two of the same sex.

Housing together & tankmates

The short version: one axolotl per tank is the safe default, and no fish at any size.

COMBINATION	VERDICT	WHY
One adult alone	BEST	No competition for food, no nipped gills, and every water reading is down to one animal. Axolotls are not social and lose nothing by living alone.
Two adults of similar size	WORKABLE	Possible with a large water volume, several hides and both animals well fed. Feed them separately, watch for nipping, and be genuinely ready to split them up.
Juveniles together	NO	Young axolotls are cannibalistic and will take each other's limbs and gills. Raise them separately.
Adult with a much smaller axolotl	NO	Anything that fits in the mouth gets swallowed, and that includes a smaller tank mate.
Any fish	NO	Small fish get eaten. Larger or nippy fish pick at exposed gills, and many carry parasites an axolotl has no defense against. Fish also want warmer water than this animal can survive.
Snails, shrimp, other amphibians	NO	Snails and shrimp are food, or a shell hazard when swallowed. Mixing amphibian species risks passing disease between them.

If you do keep two

- Match them by size, not by age. A size gap is the thing that turns cohabiting into predation.
- Give each animal at least one hide it can occupy alone, and enough floor length that they are not forced into contact.
- Target-feed each one with tongs so neither goes hungry and neither has to compete at the same spot.
- Keep a second cycled tank or a tub ready before you house them together, not after the first bite.
- Recheck the water more often. Two animals in a 40 gallon (150 L) tank produce a lot more waste than one.

NIPPED GILLS AND LIMBS REGROW

An axolotl that loses a gill filament, a toe or even a whole limb to a tank mate usually regenerates it, which is genuinely remarkable and is not a reason to tolerate the injury. Every bite is an open wound in water that carries bacteria and fungus, so separate the animals, keep the water clean, and watch the site for white growth or redness.

NEVER INTRODUCE A NEW AXOLOTL STRAIGHT INTO AN OCCUPIED TANK

A new animal goes into its own quarantine setup first, every time. See page 17.

Bringing one home, quarantine & handling

The first week, and the reason this animal stays in the water.

Acclimation, step by step

- 1. Get the animal home fast and dark. Keep the container out of the sun, and never leave it in a parked car.
- 2. Float the sealed container in the tank for 15 to 20 minutes so the temperatures equalize. Temperature is the shock that matters most here.
- 3. Add tank water to the container a little at a time over 30 to 60 minutes, until the volume has roughly doubled, so pH and hardness shift gradually.
- 4. Move the axolotl across by guiding it into a submerged cup or container. Do not tip the shop's water into your tank, and do not use a net.
- 5. Leave it alone. Dim the lights, skip the first day's meal, and let it find a hide. Most axolotls eat within a day or two of arriving.

Quarantine

If you already keep an axolotl, a new one never goes straight into the established tank. Set it up in its own small tank or tub, cycled or with daily water changes, for a few weeks while you watch it eat, move, breathe and pass waste normally. This is also the window for a first veterinary check, including a fecal test for parasites. Quarantine protects the animal you already have, and it protects the new one from an infection it picks up while still stressed from the move.

Handling: why the answer is almost always no

An axolotl has thin, scaleless, permeable skin, exposed external gills doing the actual work of breathing, and a skeleton that is largely cartilage rather than bone. There is no robust slime layer to fall back on the way a fish has. Handling risks skin damage, gill damage, stress and an infection at any point of contact, and it offers the animal nothing in return. This is a pet you watch, not one you hold.

WHEN A MOVE IS GENUINELY NECESSARY	HOW TO DO IT
Tank maintenance or a rehouse	Guide the animal into a submerged container while it stays underwater, then lift the container. Never lift the animal itself out by hand.
A vet visit or a health check	Same method, into a lidded container of its own tank water, insulated and kept cool for the trip. See page 35.
A treatment bath	Same method again, into the prepared bath at a matched temperature. See page 28.
Never	Nets. The fine gill filaments and toes snag in mesh and tear. Wash your hands before and after any contact, with no soap residue left on them.

THE RELATIONSHIP IS REAL, IT IS JUST NOT TACTILE

Axolotls learn who brings food, and most will come to the front of the tank and take a worm from tongs. That is the interaction this species has to offer, and no amount of good care turns it into a pet you can hold.

Diet & feeding by age

A carnivore that hunts by smell, swallows whole, and slows right down as it grows.

Axolotls eat invertebrate prey and need no plant matter at all. Earthworms and nightcrawlers are the staple: nutritionally complete on their own, strongly scented, and accepted by almost every axolotl. Everything else in the diet is a supplement to that or a treat alongside it.

AGE	FREQUENCY	WHAT AND HOW MUCH
Hatchling, 0 to 3 months	2 to 3 times daily	Live food, typically baby brine shrimp, then chopped blackworms. Hatchlings need moving prey.
Juvenile, 3 to 6 months	About daily	Chopped nightcrawler and small soft sinking pellets. Bare-bottom tank at this size.
Sub-adult, 6 to 12 months	Every 1 to 3 days	Cut worm and pellets, sized to the animal rather than to the packet.
Adult, 12 months and over	Every 2 to 3 days	Whole or halved nightcrawler, or roughly five soft pellets. Adults digest slowly in cold water and genuinely need the gap.

Portion size, and how to tell

- Offer only what the animal finishes in two to three minutes, and no piece wider than its own head.
- Remove anything uneaten within 15 to 30 minutes. Leftover worm is the fastest route to an ammonia spike.
- Target-feed with soft tongs, held just in front of the animal. It hunts by scent and water movement, not sight, so food dropped elsewhere in the tank is often food wasted and fouling the water.
- Feed each animal separately if you keep two, so neither goes short and neither strikes at the other in the confusion.

THE MOST COMMON FEEDING MISTAKE IS NOT WHAT YOU FEED

It is feeding an adult on a juvenile's schedule. Daily food for a grown axolotl produces an obese animal and a tank fighting its own waste. Obesity is a recognized problem in this species, and unlike most axolotl problems it builds so slowly that nobody notices until the body is visibly wider than the head.

APPETITE IS YOUR BEST DAILY HEALTH SIGNAL

A steady feeding response, every 2 to 3 days for an adult, is the simplest vital sign you have and it takes no equipment. Log it on page 37. Refusal is rarely the first thing to go wrong, but it is very often the first thing you notice.

Staples, treats & the never-feed list

A short shopping list, and a shorter list of things that cause real harm.

FOOD	TIER	NOTES
Earthworms and nightcrawlers	STAPLE	Nutritionally complete and readily taken. Bait-shop or pesticide-free garden sourced. Cut to size for juveniles. Rinse before offering.
Soft sinking pellets for axolotls or carnivorous amphibians	STAPLE	A practical primary or supplementary diet. Salmon-based and similar soft pellets are standard in research colonies. About five per adult feeding.
Blackworms, live or frozen	OCCASIONAL	Excellent for juveniles and for tempting a fussy eater. Rinse frozen portions and thaw before offering.
Frozen bloodworms, brine shrimp, daphnia	OCCASIONAL	Good variety and good for hatchlings. Too thin nutritionally to be the whole diet for an adult.
Feeder fish, live or frozen	NEVER	Real parasite risk, and many common feeder species carry an enzyme that causes vitamin B1 deficiency over time.
Beef heart, chicken and other mammal meat	NEVER	Axolotls cannot properly digest mammalian fat and protein.
Adult mealworms, crickets, crayfish and other hard-shelled prey	NEVER	Exoskeletons are hard to digest and can injure the gut on the way through.
Anything wider than the animal's head	NEVER	It gets swallowed anyway. Oversized food is one of the two routes to an impaction, the other being the tank floor.

Supplements

None are needed. A diet of worms plus a formulated pellet covers this species' requirements without added vitamins or minerals, which is unusual among exotic pets and one less thing to get wrong. There is no calcium dusting step here, no D3 schedule, and no multivitamin.

STORE FOOD PROPERLY, AND BUY SMALL

Keep nightcrawlers refrigerated in their bedding and check them before every feed, discarding any that are dead or smell sour. Keep frozen foods frozen, thaw only the portion you are about to use, and never refreeze. Pellets go stale: buy a small tub, keep it sealed and dry, and replace it every few months rather than working through a bulk bag for a year.

THE WORM OBJECTION

Plenty of people would rather not keep a tub of live worms in the fridge. That is a fair preference and it is worth knowing before you buy the animal, because the alternative is a pellet-only diet, which works but is a narrower one. If worms are a hard no, at least buy the best pellet you can and add frozen blackworms for variety.

Common mistakes

Ranked by how much harm each one actually does.

MISTAKE	COST	THE FIX
Letting the water run warm	Fatal	The single biggest killer of pet axolotls, and the most common. Plan cooling before summer, not during it. Page 8.
Gravel or small stones on the floor	Fatal	Swallowed while feeding, then a blockage that needs surgery. Bare bottom or fine sand under 1 mm. Page 14.
Adding the animal before the tank is cycled	Severe	Ammonia and nitrite burns from day one. Cycle first, and if you are already past that point, page 11 has the recovery plan.
A heater in the tank	Severe	There is no scenario in which an axolotl tank needs one. If the tank came as a kit, take the heater out.
Fish as tankmates	Severe	Nipped gills, stolen food, parasites, and a temperature conflict with no solution. Page 16.
Too much filter flow	Chronic stress	Baffle the output or switch to a sponge filter. Gills held flat and swept back are the tell. Page 10.
Feeding an adult daily	Chronic	Obesity and a fouled tank. Every 2 to 3 days. Page 18.
Bright lighting	Chronic stress	No eyelids, no escape. Dim, indirect light and a shaded hide.
Handling for fun	Injury risk	Skin and gill damage, and an infection route. Medical necessity only. Page 17.
A tank with no lid	Escape	Axolotls get themselves out of open tanks and dry out fast on the floor.
Treating symptoms instead of testing	Wasted days	Reaching for medication before a thermometer and a test kit is how a temperature problem gets treated as a disease for a week.

THE PATTERN BEHIND ALMOST ALL OF THEM

Nine of the eleven above are environment, not disease. Axolotls are hardy animals in the right conditions and fragile ones outside them, which is why this package spends most of its pages on the tank rather than on medicine. Get the temperature, the floor and the cycle right and the health section becomes something you read once and rarely need.

Enrichment: hides, foraging & novelty

Not toys. Choice: somewhere to rest, cover that feels secure, and food that takes some finding.

THE BAR FOR "SAFE" IS HIGHER HERE

No eyelids, delicate external gills and a real impaction risk from anything loose mean every item in this tank has to be smooth, stable and either larger than the head or fine enough to pass. That rules out a lot of what an aquarium shop sells as decor.

Structure and cover

Two hides per adult is the floor, three or four of different shapes and heights is better, because a hide is how an axolotl controls its own light level and its distance from anything alarming. Sideways terracotta pots with the drainage hole sealed, wide sanded PVC sections and smooth resin caves all work. Live or silk plants add cover: java fern and anubias tied to decor rather than buried, moss balls kept larger than the animal's head. A gentle current on one side of the tank paired with genuinely still water on the other lets the animal choose its own water movement instead of living in whatever you picked.

Foraging

Dropping food in the same spot and walking away skips the hunting behavior entirely. Tong-feeding, or scattering pieces of nightcrawler so the animal has to locate them by scent, is a real upgrade and costs nothing. A sand saucer, a shallow dish of sub-1 mm sand with small food items hidden in it, adds searching behavior in a contained way without loose sand across the whole floor. Rotate where you feed every few days.

Sensory novelty

A mild monthly rearrange of the non-critical decor, the plants, the extra tunnel, the spare stone, adds spatial novelty without taking away the hides the animal actually relies on. Introducing one new safe item for five to seven days and then rotating it out works the same way. Floating plants or a dimmed light create the dappled shade this species is drawn to. Strong current and bright light are not enrichment for an axolotl, they are just stress with a different name.

ENRICHMENT	PRIORITY	WHY
Two or more hides	Essential	Security and light control, for an animal that cannot close its eyes.
Dim, indirect lighting	Essential	A genuine stress reducer for this species, not a comfort preference.
A still-water zone	Essential	Lets the animal opt out of the current entirely.
Scatter and tong feeding	High	Turns a 20 second event into real hunting behavior.
Plants and floating cover	Medium	Shade, texture, and something to rest against.
Monthly rearrange, rotating item	Medium	Novelty without removing anything the animal depends on.

Growth, body condition & reading waste

Three checks you can do through the glass, with nothing in your hands.

Growth reference

AGE	TYPICAL LENGTH	WHAT CHANGES AT THIS STAGE
3 to 6 months	2 to 5 in (5 to 13 cm)	Moving onto chopped worm and pellets. Still bare bottom.
6 to 12 months	5 to 8 in (13 to 20 cm)	Feeding slows toward every 1 to 3 days. Fine sand becomes an option toward the top of this range.
12 months and over	9 to 10 in (23 to 25 cm) typical, sometimes to about 12 in (30 cm)	Adult schedule, every 2 to 3 days. Sexually mature at around a year.

Growth is not a race, and a cold tank grows an axolotl more slowly than a warm one. That is normal and preferable. An animal that stalls completely, or thins while eating, is a vet conversation rather than a feeding adjustment.

Body condition, viewed from above

CONDITION	WHAT YOU SEE LOOKING DOWN THROUGH THE WATER
Underweight	Body noticeably narrower than the head, a visible taper behind the gills, the outline of the spine or hips. Investigate feeding and water first.
Good	Body roughly as wide as the head, with smooth, even sides from gills to hips.
Overweight	Body clearly wider than the head, a rounded bulge behind the front legs, and a sagging profile from the side. Stretch the feeding interval, do not fast the animal outright.
Bloated, not fat	Sudden swelling, a tight or shiny look, often with floating. That is a health problem and not a diet one. Page 26.

Reading waste

WHAT YOU FIND	WHAT IT MEANS
Firm, dark, sausage-shaped	Normal. Siphon it out at the next opportunity, since it is an ammonia source sitting on the floor.
Stringy, pale or mucus-heavy	Digestive upset. Check the temperature first, then what and how much was last fed.
Sand or grit visibly in it	The animal is ingesting substrate. Small amounts of fine sand pass, which is the point of fine sand, but take it as a warning to check depth and grain size.
Nothing at all for several days after eating	Possible impaction, particularly with bloating or floating. Page 26.

PHOTOGRAPH THE ANIMAL MONTHLY

One photo from directly above, same spot, same light, every month. Slow changes in width, gill length and color are invisible day to day and obvious across four photos, and it is the most useful thing you can hand a vet.

Health red flags

What to watch for, when to call, and what to have ready when you do.

CALL AN AMPHIBIAN OR AQUATIC VET PROMPTLY IF YOU SEE

White or grey cottony growth on skin or gills that has not improved within 48 hours of correcting the water · suspected impaction, especially bloating, floating or no waste passed · red streaking, open wounds or ulcers · appetite loss lasting 5 days or more · bulging eyes · sudden severe floating or loss of balance · any injury with visible tissue damage

Test before you treat

Almost every condition in this section traces back to temperature or water quality, so the first two actions for any symptom are a thermometer reading and a full test. A vet will ask for both before anything else, and a course of medication into a tank that is running at 75°F (24°C) fixes nothing. Correcting the water is not a delay in treatment, it is usually the largest part of the treatment.

What to have ready when you call

- ☐ Current water temperature, and how long it has been out of range
- ☐ Ammonia, nitrite, nitrate and pH readings, with the date tested
- ☐ Tank volume, substrate type, filter type and how long the tank has been running
- ☐ What you feed, how much and how often, and the date of the last meal
- ☐ When symptoms started and whether they are worsening
- ☐ Photos from above and from the side, plus the Owner Log from page 37

Normal against concerning

NORMAL	CONCERNING
Sitting still for long stretches. This is an ambush animal, not a busy one	Lethargy with no response to food or a shadow passing
Gills fanning, twitching or flicking periodically	Gills held curled forward, or shrinking over weeks
An occasional gulp of air at the surface, then back down	Stuck at the surface, or unable to sink
Skipping one meal	Refusing food for 5 days or more
Slower feeding and less movement in genuinely cold water	The same in water at normal temperature
Shedding a wisp of skin or slime occasionally	Cloudy or peeling patches of skin, or a film over the body
Colors deepening or changing subtly as the animal matures	New red, inflamed or ulcerated patches

Heat stress & reading the gills

The condition behind most of this section, and the early warning system on the animal's own head.

Cause: water above 72°F (22°C) for any sustained period, and acutely above 75°F (24°C). **Signs:** gill tips curling forward toward the face, appetite loss, restlessness or the opposite, uncontrolled floating, pale patches, and a general fragility that shows up as a fungal or bacterial infection a week later. **Response:** bring the temperature down, steadily rather than instantly, and test the water while you do it.

COOL IT DOWN, BUT NOT ALL AT ONCE

Aim to lose no more than 1 to 2°F (about 1°C) an hour. Move the tank out of sun, get a fan across the surface, float sealed bottles of frozen dechlorinated water and swap them as they melt, cut the lights, and stop feeding until the temperature is back in range. Never tip loose ice into the tank, and never chase a fast drop with a large cold water change.

Reading the gills

WHAT THE GILLS LOOK LIKE	WHAT IT USUALLY MEANS
Long filaments, fanned outward and back, deep pink or red	Healthy and well oxygenated. This is the baseline to photograph so you have something to compare against.
Curled forward toward the face	Stress, most often heat or water quality. The earliest visible sign there is, and often days ahead of appetite loss.
Pale or washed out	Poor oxygenation or poor condition. Check temperature and surface movement.
Very dark red and flared	Working hard for oxygen. Warm water holds less of it, so check the thermometer before anything else.
Filaments short, stubby or thinning out	Longer-running water quality problems, excessive flow, or nipping from a tank mate. They regrow once the cause is removed.
White cottony growth on the filaments	Fungal infection taking hold on stressed tissue. Page 25.

WHY GILLS ARE SUCH A GOOD EARLY WARNING

They are external, well supplied with blood, and the first tissue affected when oxygen drops or the water turns hostile. That means they change before appetite does and long before anything visibly fails. Look at them for five seconds every day, in the same light, and you get days of notice on almost every problem in this section.

COLD IS NOT THE MIRROR IMAGE OF HEAT

A tank that drifts down to the low 50s°F (around 11 to 12°C) makes an axolotl sluggish and puts it off food, and that is genuinely less dangerous than the same drift upward. Vets deliberately hold sick axolotls in cool water during treatment. Correct a cold tank, but do not panic about it the way you would about a warm one.

Fungal & bacterial infection

The two conditions that follow warm water and dirty water around, and how to tell them apart.

Fungal infection

Signs: white or grey cotton-like tufts growing from the skin or gill filaments, most often starting at a site that was already damaged. **Cause:** almost always secondary. Warm water, poor water quality, stress or an existing injury opens the door, and the fungus, usually a *Saprolegnia* species, walks through it. **Response:** correct the water first, cool the tank into the lower half of the range, and remove whatever caused the original injury. Salt baths at the dose on page 28 are the standard next step. A case that has not improved within 48 hours of the water being right, or one spreading over the body, needs a vet and a prescribed antifungal.

Bacterial infection

Signs: red inflamed patches, red streaking through the skin or into the limbs, open sores or ulcers, swelling, and an animal that is lethargic and off its food. Advanced septicemia may show as bleeding under the skin or a body-wide redness. **Cause:** commonly *Aeromonas* and *Pseudomonas* species, which are present in any tank and become a problem when water quality slips or the animal is already weakened. **Response:** this is a vet case, not a home one. Caught early, systemic antibiotics are usually curative. Left to advance, the prognosis is poor.

TELLING THEM APART	FUNGAL	BACTERIAL
What you see	Fluffy white or grey tufts standing off the surface	Redness, streaking, ulcers, swelling, no fluff
Where it starts	Gills or an existing wound	Limbs, belly and vent, or a wound
Speed	Days, and visible from the outset	Can move fast, and the animal often looks unwell before the skin does
First response	Fix water, cool, salt bath, vet if not improving in 48 hours	Fix water and call a vet the same day

WHAT A VET MAY REACH FOR

Published veterinary guidance for axolotls uses enrofloxacin, gentamicin or amikacin at 5 mg per kg by injection, sometimes with the animal held in cool water at 41 to 59°F (5 to 15°C) for several weeks during treatment, and notes that tetracyclines are unsafe in this species. This is here so you can discuss it with a vet, not so you can dose an animal at home. Antibiotic dosing in amphibians is weight-based and route-specific, and getting it wrong causes its own harm.

PREVENTION REALLY IS THE TREATMENT

Both of these are opportunists. An axolotl in cold, clean, cycled water with no injuries and no tank mate biting it rarely gets either, which is why the highest-leverage page in this whole health section is the thermometer on page 8.

Impaction, floating & gas

The preventable emergency, and how to tell it from the harmless version of the same symptom.

Cause: a swallowed object that cannot pass. Gravel is the classic, oversized food is the other, and both come straight from the way this animal feeds: it inhales whatever is in front of its mouth. **Signs:** bloating, floating or an inability to sink, appetite loss, no waste passed for days, and lethargy. **Response:** stop feeding and call a vet. A gastric foreign body can sometimes be retrieved from the stomach by a vet with lubricated forceps, and one that has moved further along needs surgery.

DO NOT WAIT TO SEE WHETHER IT RESOLVES

An impaction that is going to pass usually does so within a couple of days. Every day past that is lost time. If you know the animal ingested substrate or an oversized item and it has stopped eating, that is a call today, not a wait-and-see.

Floating: three different problems that look alike

PRESENTATION	LIKELY CAUSE	WHAT TO DO
Rises to the surface, gulps air, swims back down	Normal behavior	Nothing. Axolotls take surface gulps.
Floats tail-up but can still swim down with effort, eating normally	Trapped gas, often swallowed air or a large meal	Usually resolves in a day or two. Fast the animal, check the temperature, and reduce anything blowing bubbles into the water. A brief spell in a cool, shallow tub can help.
Cannot sink at all, or lists to one side, with bloating	Impaction or serious gas buildup	Vet, promptly. Do not feed.
Floating with heat signs, curled gills, in a warm tank	Heat stress	Cool the water gradually and reassess. Page 24.

Preventing it entirely

- Bare bottom for anything under about 5 in (13 cm), and fine sand under 1 mm at most for adults.
- No food wider than the animal's head, ever.
- Target-feed with tongs so the animal strikes at the food rather than at the floor around it.
- Keep sand shallow, and siphon it at every water change so there is less loose material to inhale.
- Remove small decor, and apply the head test on page 14 to anything new.

WHY THIS PAGE EXISTS AT ALL

Impaction is almost entirely a husbandry outcome. Bare bottom or fine sand, correctly sized food, and it more or less stops being a risk. Gravel in an axolotl tank is the one decor choice in this package that has a body count.

Burns, injuries & other conditions

The rest of the list, most of which is water chemistry wearing a different face.

Ammonia and nitrite burns

Signs: red, inflamed patches on the skin or gills, gills held curled, appetite loss, and an animal that looks generally irritated. **Cause:** an uncycled tank, a crashed filter, or a missed run of water changes. **Response:** test, then change 25 to 50 percent of the water with dechlorinated, temperature-matched water, and repeat daily until both readings hold at 0. A conditioner that temporarily detoxifies ammonia and nitrite buys time between changes. Damaged tissue recovers once the water is right, but broken skin is also an entry point for the infections on page 25, so keep watching the site.

Injuries and regeneration

Nipped gill filaments, lost toes and even whole limbs regrow, which is genuinely extraordinary and is not a reason to be relaxed about the wound. Regeneration takes weeks and it happens in water that carries bacteria and fungus, so the job is keeping the site clean while the animal does the rest: remove the cause first, whether that is a tank mate or a sharp edge, keep water quality immaculate, keep the tank in the cooler half of the range, and watch daily for white growth or spreading redness. A wound that looks worse rather than better after a few days is a vet call.

CONDITION	SIGNS	CAUSE AND RESPONSE
Exophthalmia, bulging eyes	One or both eyes protruding	Associated with prolonged high nitrate. Bring nitrate down with water changes and keep it under 10 ppm. Vet if it persists.
Obesity	Body clearly wider than the head, sagging sides	Overfeeding, usually an adult still on a juvenile schedule. Stretch the interval, do not starve. Page 18.
Anorexia, refusing food	No feeding response for 5 days or more	Check temperature and water first, then look for impaction or infection. A vet visit for anything past a week.
Internal parasites	Weight loss despite eating, abnormal waste, lethargy	Uncommon in captive-bred stock. Diagnosed by a fecal test, treated by a vet. Worth screening a new animal during quarantine.
Chemical or soap exposure	Sudden distress, mucus, irritation	Permeable skin absorbs everything. Large water change now, and never rinse tank equipment with detergent.

DO NOT MEDICATE A TANK ON A HUNCH

Amphibian skin absorbs what is in the water, so an off-the-shelf fish treatment is not a low-risk experiment. Several common aquarium medications are unsuitable here, and dosing the display tank can wipe out the bacteria colony too. Correct the water, use a salt bath if indicated, and get a diagnosis first.

Tubbing, cooling & salt baths

Three interventions worth knowing before you need them, and where each one stops.

Tubbing

Tubbing means housing the axolotl in a plain container of dechlorinated, temperature-matched water changed completely every day, instead of in the tank. No filter and no cycle to maintain, because the daily change does that job. It is the standard answer while a tank cycles with the animal already in it, while an injured animal heals away from a tank mate, during quarantine, and whenever water quality has gone wrong faster than water changes can fix it.

- A clean, food-safe plastic tub with a lid, big enough to turn around in and deep enough to cover the animal.
- Change 100 percent of the water daily, matching temperature. Keep the tub cool and dark.
- Never use soap or detergent on it, and keep the tub for this purpose only.
- Feed as normal, removing uneaten food within 15 to 30 minutes.

Cooling as treatment

Cooler water slows bacterial growth and lowers the animal's metabolic demand, which is why published veterinary guidance for treating bacterial infection in axolotls holds them at 41 to 59°F (5 to 15°C) for several weeks during the course. Hobbyists call the home version fridging. Be precise about what it is: a therapeutic measure, under veterinary direction, in daily-changed tubbed water, brought down gradually rather than by moving an animal from a warm tank into a cold fridge. It is not routine husbandry and not a substitute for fixing the tank.

Salt baths

ELEMENT	FIGURE
Concentration	10 to 15 grams of salt per liter of water
Duration	10 minutes per bath
Frequency	Once or twice daily, for as long as a vet directs
Salt type	Non-iodized aquarium or sea salt, with no anti-caking additives
Water	Dechlorinated and matched to the animal's current temperature
Used for	Fungal infection, alongside correcting the water. Not a general tonic

Dissolve the salt fully before the animal goes in, never over it. Stay for the full ten minutes, end early if it shows real distress, then return it to clean, cool tubbed water. Excessive salt treatment does harm, so more and longer is not better, and salt does not belong in the display tank as a preventive.

WHERE HOME CARE STOPS

These three are supportive. They buy time and close the door on whatever caused the problem. They do not treat a bacterial septicemia, a blockage or an ulcer. If a fungal case has not turned around within 48 hours of the water being right, or you see red streaking, open wounds, bulging eyes or a refusal to eat past five days, stop managing it at home.

Setup checklist & targets

Equipment

- ☐ 20 gal long (78 L) tank minimum for one adult, 40 gal breeder for two
- ☐ Secure glass or mesh lid
- ☐ Sponge filter, or a canister or hang-on-back with the output baffled
- ☐ Cooling plan: chiller, room air conditioning, or a fan and a genuinely cool room
- ☐ Digital thermometer reading in the water, not stuck to the glass
- ☐ Liquid test kit for ammonia, nitrite, nitrate and pH
- ☐ Bare bottom, or fine sand under 1 mm for an adult
- ☐ Two or more smooth hides per animal
- ☐ Dechlorinator, and a siphon and dedicated buckets
- ☐ Soft feeding tongs
- ☐ A spare lidded tub for tubbing, quarantine or transport
- ☐ Nightcrawler supply and a soft sinking pellet, sourced before day one

Targets

TARGET	RANGE
Water temperature	60 to 68°F (16 to 20°C), cooler half preferred. Never above 72°F (22°C). Above 75°F (24°C) illness is likely
Ammonia	0 ppm
Nitrite	0 ppm
Nitrate	Under 10 ppm
pH	7.4 to 7.6, and stable, inside a tolerable 6.5 to 8.0
Hardness	Moderately hard. Never distilled, RO or very soft water without remineralizing
Water change	20 to 30 percent weekly, temperature-matched and dechlorinated
Feeding	Adults every 2 to 3 days, juveniles daily, uneaten food out within 15 to 30 minutes

Before the axolotl comes home

- ☐ Tank fully cycled: ammonia 0 and nitrite 0 twenty four hours after dosing ammonia
- ☐ Nitrate brought under 10 ppm by a large water change
- ☐ Temperature held in range across a full warm day, not just overnight
- ☐ Cooling method tested under the worst conditions you expect
- ☐ Amphibian or aquatic vet found, with the number written down
- ☐ Legal status confirmed for your state and city. Page 6

Emergency & quick targets

CALL AN AMPHIBIAN OR AQUATIC VET NOW IF

White or grey cottony growth not improving 48 hours after the water is corrected · bloating, floating, or no waste passed for days · red streaking, open wounds or ulcers · bulging eyes · refusing food for 5 days or more · loss of balance · known ingestion of substrate or oversized food

Vet _____ Tank volume _____
 Emergency vet _____ Date set up _____
 Backup contact _____ Test kit expires _____

READING	TARGET
Water temperature	60 to 68°F (16 to 20°C), never above 72°F (22°C)
Ammonia	0 ppm
Nitrite	0 ppm
Nitrate	Under 10 ppm
pH	7.4 to 7.6
Feeding	Adults every 2 to 3 days, juveniles daily
Water change	20 to 30 percent weekly
Salt bath, if directed	10 to 15 g per liter, 10 minutes, once or twice daily

The water is too warm, right now

1. Lights off, lid open, fan across the water surface.
2. Float sealed bottles of frozen dechlorinated water and swap as they melt. No loose ice.
3. Aim for no more than 1 to 2°F (about 1°C) an hour of change.
4. Stop feeding until the temperature is back in range.
5. Test ammonia and nitrite: warm water and a stressed filter often arrive together.

HAVE THIS READY WHEN YOU CALL

Water temperature and how long it has been out of range, ammonia, nitrite, nitrate and pH with the test date, tank volume and substrate, what and when you last fed, when symptoms started, and a photo from above.

Budget & shopping list

Real line items that add up to the totals, rather than a headline number.

ITEM	COST
Tank, 20 gal long (78 L) or larger, with lid	\$60 to \$170
Filtration: sponge filter, or canister or hang-on-back plus a baffle or spray bar	\$20 to \$120
Substrate: fine sand under 1 mm, or nothing at all for bare bottom	\$0 to \$25
Hides, plants and decor	\$20 to \$50
Digital thermometer	\$10 to \$15
Liquid test kit: ammonia, nitrite, nitrate, pH	\$15 to \$25
Dechlorinator	\$8 to \$15
Siphon, dedicated buckets, feeding tongs, spare lidded tub	\$25 to \$45
Subtotal, before cooling	\$158 to \$465
Cooling option A: fan and a genuinely cool room	\$15 to \$40
Cooling option B: aquarium chiller	\$150 to \$400
The axolotl itself, standard morph	\$30 to \$100
Total with fan cooling, including the animal	\$203 to \$605
Total with a chiller, including the animal	\$338 to \$965

Monthly running cost

ITEM	COST
Food: nightcrawlers, pellets, frozen variety	\$5 to \$15
Dechlorinator and test kit reagents	\$3 to \$8
Filter media, spread across the year	\$2 to \$5
Electricity, mostly the chiller and mostly in summer	\$5 to \$17
Monthly total	\$15 to \$45

WHERE NOT TO SAVE

Cooling, and the test kit. Cooling is the line tied to the leading cause of death in pet axolotls, and the test kit is the only way to see the second one coming. Save on decor and buy the tank used instead.

TWO COSTS THAT ARE NOT ON THIS PAGE

Veterinary care, which is not predictable and is worth a small standing fund, and electricity in a hot climate, where a chiller running most of the year pushes past the range above. Check your chiller's rated wattage against your own electricity price rather than an average.

First 30 days

Assumes the tank was cycled before the animal arrived. If it was not, work page 11 alongside this.

Week 1

- ☐ Acclimate slowly and then leave the animal alone. Dim lights, minimal traffic past the tank
- ☐ Check the temperature at the warmest part of the day, every day
- ☐ Test ammonia and nitrite daily for the first week, whatever the tank did before
- ☐ Offer food from day two. Do not worry about one or two refusals while it settles
- ☐ Photograph the animal from above, as the baseline for every later comparison

Week 2

- ☐ Settle into the feeding schedule for its age, and note how long it takes to find the food
- ☐ First full test and water change on a fixed day of the week
- ☐ Start the Owner Log on page 37, one row a week
- ☐ Confirm the filter flow is not pushing the animal around, and baffle it if it is

Week 3

- ☐ Look at the gills in good light and compare against your week 1 photo
- ☐ Add or move a hide if the animal only ever uses one spot
- ☐ Try tong feeding and one scatter feed, from page 21
- ☐ Call the vet you found and ask whether they see axolotls, before you ever need them

Week 4

- ☐ Review four weeks of the log for trends rather than single readings
- ☐ Check the substrate: siphon it, confirm the depth, and look for grit in the waste
- ☐ Second photo from above, and compare body width against head width
- ☐ Plan for the next season: whatever holds temperature now may not in July

Days 31 to 90

- ☐ Keep testing weekly even once everything looks stable. Stability is the thing you are measuring
- ☐ Rotate one enrichment item a month, and rearrange the non-critical decor
- ☐ Reassess substrate as the animal grows past 5 in (13 cm)
- ☐ Fill in the pet-sitter sheet on page 36 before you need it, not the night before a trip

Symptom quick reference

Start every row by reading the thermometer and testing the water. It is the answer more often than not.

SYMPTOM	LIKELY CAUSE	ACTION
Gills curled forward toward the face	Heat or water quality stress	Test and check temperature, correct gradually. Page 24
Gills shortening or thinning over weeks	Water quality, too much flow, or nipping	Test, baffle the filter, separate animals. Pages 10 and 16
White or grey cottony growth	Fungal infection on stressed tissue	Correct water, cool, salt bath, vet at 48 hours. Pages 25 and 28
Red inflamed patches on skin or gills	Ammonia or nitrite burn	Test, 25 to 50 percent water change, repeat daily. Page 27
Red streaking, ulcers or open sores	Bacterial infection	Vet the same day. Page 25
Floating and unable to sink, with bloating	Impaction or serious gas	Stop feeding, vet promptly. Page 26
Floating tail-up but still eating	Trapped gas	Fast a day or two, check temperature, watch. Page 26
No waste passed for several days after eating	Possible impaction	Stop feeding, vet promptly. Page 26
Refusing food 5 days or more	Temperature, water, impaction or infection	Test first, then vet. Page 23
Bulging eyes	Prolonged high nitrate	Water changes to bring nitrate under 10 ppm, vet if persistent. Page 27
Body visibly wider than the head	Obesity from overfeeding	Stretch the feeding interval, do not fast. Page 18
Ammonia or nitrite above 0 ppm	Uncycled tank or a stalled filter	Water change now, then daily. Page 11
Nitrate above 20 ppm	Water change overdue, or overfeeding	Larger or more frequent changes. Page 13
Temperature above 72°F (22°C)	Cooling not keeping up	Emergency cooling steps on page 30
Cloudy water or a sour smell	Uneaten food, or a filter that has stopped	Find the cause today, siphon waste, check the filter. Page 13
Missing toes, gill filaments or a limb	A tank mate, or an injury on decor	Separate, remove the hazard, keep water immaculate. Page 27
Animal out of the tank	Lid not secure	Return it to water immediately, secure the lid, watch for skin damage

Daily, weekly & seasonal routine

About twenty minutes a week, plus a minute a day.

Daily

- ☐ Read the water temperature
- ☐ Look at the gills for five seconds in good light
- ☐ Check the filter is running and the surface is moving
- ☐ Feed on schedule, and remove leftovers within 15 to 30 minutes
- ☐ Siphon out any visible waste if it is easy to reach

Weekly

- ☐ Test ammonia, nitrite, nitrate and pH, before the water change
- ☐ Change 20 to 30 percent, dechlorinated and temperature-matched
- ☐ Siphon the substrate as you go
- ☐ Wipe the glass and check decor for new sharp edges
- ☐ Write one row in the Owner Log on page 37

Monthly

- ☐ Rinse mechanical media in tank water, never tap water
- ☐ Photograph the animal from above and compare against last month
- ☐ Read back through the log for slow trends
- ☐ Rearrange one piece of non-critical decor, rotate one enrichment item
- ☐ Check the chiller or fan is still doing what it did last month

Seasonal

- ☐ Before summer: test the cooling setup against a genuinely hot day, not an average one
- ☐ Before summer: buy or check the backup, frozen bottles and a battery air pump
- ☐ In winter: confirm the room has not gone so cold that the tank sits below the range
- ☐ Twice a year: check the test kit expiry date and replace reagents
- ☐ Annually: a wellness check with the vet, and update the pet-sitter sheet

FIX THE DAY, NOT THE INTENTION

Pick one day a week and always test and change water on it. A routine tied to a day survives a busy month. A routine tied to when the tank looks like it needs it drifts, and by the time an axolotl tank looks like it needs attention the readings moved days ago.

WHAT IS NOT ON THIS LIST

Handling, deep cleans, and medication. None of them belong in a routine. If you find yourself doing any of the three regularly, something upstream in the setup needs fixing instead.

Power outages, heat waves & transport

For an axolotl, a power cut is a temperature problem and an oxygen problem at the same time.

When the power stops, three things stop with it: the chiller, the filter's flow, and the gas exchange that flow provides. Warm water holds less dissolved oxygen at exactly the moment the water is warming up, and the bacteria colony starts dying within hours without flow. Unlike a tropical fish keeper, you are not holding heat in. You are keeping it out.

Do

- Insulate: close the blinds, cover the tank sides with a blanket or foam, keep the room dark and shut.
- Float sealed bottles of frozen dechlorinated water, swapping them as they melt, aiming for a slow hold rather than a fast drop.
- Run a battery air pump into a sponge filter or air stone. That covers oxygen and keeps the media alive.
- Stop feeding. No filter means no processing, and uneaten food consumes oxygen while producing ammonia.
- If it runs long, move the biological media into an aerated container or the tank so the colony survives.

Don't

- Don't tip loose ice into the tank, and don't chase a fast temperature drop. Aim for no more than 1 to 2°F (about 1°C) an hour.
- Don't restart a filter that has been off for hours without rinsing the media. Stagnant media dumps decomposed waste into the tank.
- Don't do a large water change in the dark with untested, unconditioned water.
- Don't use candles, a gas stove or any open flame near the tank.

Leaving town

TRIP LENGTH	WHAT IS ACTUALLY NEEDED
1 to 3 nights	Nothing much. A healthy adult fasts comfortably for a few days. Water change and a feed before you go, and no automatic feeder.
4 to 7 nights	A visit every two or three days to check temperature and filter, with food pre-portioned so it cannot be overfed.
Over a week	A sitter doing real visits, plus a water change if the trip runs long. Walk them through page 36 in person first. In summer, never leave a chiller-dependent tank unattended for a week.

Transport and vet trips

- Move the animal in a lidded container of its own tank water, inside an insulated cooler so it stays dark and cool.
- Do not feed for 24 hours before a move. Less waste, and less chance of a regurgitated meal.
- Guide the animal into the container underwater. No nets, no lifting it by hand.
- Never leave the container in a parked car at any time of year.
- On arrival, acclimate to temperature before anything else, as on page 17.

FILL IN AND LEAVE BY THE TANK

Pet-sitter sheet

Everything a sitter needs on one page. Walk them through it once in person before the first solo visit.

Animal's name and color _____

Vet _____

Tank volume and location _____

Backup contact _____

My phone _____

Feeding days _____

THE ONE THING THAT MATTERS MOST

This animal must stay cold. The thermometer should read between 60 and 68°F (16 and 20°C). If it reads above 72°F (22°C), that is an emergency: close the blinds, turn the lights off, put a fan across the water surface, float a sealed bottle of frozen water from the freezer, and call me. Never add a heater, and never add ice cubes directly to the tank.

Each visit

- | | |
|---|--|
| ☐ Read the thermometer and write the number down | ☐ Feed ONE pre-portioned container, only on the days listed above |
| ☐ Check the filter is running and the water surface is moving | ☐ Scoop out anything not eaten after 20 minutes |
| ☐ Look at the animal: is it upright on the bottom, and are the gills fluffy rather than curled forward | ☐ Wash your hands afterward, and don't rinse anything in the kitchen sink |

Please don't

- Feed extra, or on a day that is not listed. Adults eat every 2 to 3 days and always look interested.
- Touch, lift or net the animal for any reason, or tap the glass. Its skin and gills damage very easily.
- Top the tank up with tap water, clean the filter, change the water, or add anything to it.
- Turn a light on above the tank or open the blinds. This animal has no eyelids and the tank must stay cool.

CALL ME IF

The thermometer reads above 72°F (22°C) · the filter is silent · the animal is floating, upside down, or has not moved between two visits · there is white fluff or a red patch on it · the water has turned cloudy or smells bad · the power has been out more than a few hours. If I don't answer, call the vet number above.

SECTION 05 • OWNER TOOLS

Owner log

One row a week, filled in on test day. Photocopy this page, or track digitally. Consistency matters more than the format.

WHY BOTHER LOGGING

The water readings are this animal's vital signs, and unlike a weight they take two minutes and no handling at all. Almost every condition in this package shows up in the numbers, or in the gills, before it is visible anywhere else. Sixteen rows is four months, long enough to see a real trend rather than a bad day, and it is the single most useful thing you can hand a vet.

Equipment & maintenance log

Equipment log

ITEM	INSTALLED	REPLACE OR CHECK BY	DONE
Tank, volume and date set up			
Filter, model and how the output is baffled			
Filter media last rinsed or replaced			
Chiller or cooling fan, model and setting			
Test kit, with expiry date			
Battery air pump and spare batteries			

Twelve-month planner

Tick the box in the month you did it. Start with whichever month you're in.

TASK	M1	M2	M3	M4	M5	M6	M7	M8	M9	M10	M11	M12
Rinse mechanical media	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Photograph from above	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐	☐
Deep siphon the substrate	☐		☐		☐		☐		☐		☐	
Test the cooling setup	☐			☐			☐			☐		
Replace test kit reagents	☐					☐						☐
Vet wellness check	☐											☐

Incident & treatment log

Record every salt bath, medication and water emergency. Doses accumulate, and a vet will ask.

DATE	PROBLEM AND READINGS	WHAT WAS DONE, AND HOW MUCH	OUTCOME

Enrichment checklist & log

Setup

- ☐ Two or more hides per animal, different shapes and heights
- ☐ A still-water zone away from the filter output
- ☐ Dim, indirect light, with shade available under plants or decor
- ☐ Open floor to walk on, with nothing sharp at ground level
- ☐ Feeding tongs, and a sand saucer if you use one

Monthly review

- ☐ Rearrange one piece of non-critical decor
- ☐ Rotate one new safe item in for 5 to 7 days
- ☐ Vary feeding: tongs, scatter, and a different spot each time
- ☐ Check every item still passes the head test on page 14
- ☐ Confirm the hides still fit the animal's current size

Enrichment log

DATE	WHAT YOU TRIED	HOW IT WENT, AND WHETHER TO REPEAT

ENRICHMENT IS NOT EXTRA FOOD

Every foraging idea in this package uses the meal the animal was already getting, presented differently. An adult still eats every 2 to 3 days whether the worm arrives on tongs, scattered, or hidden in a sand saucer. Adding a session means dividing the same ration, never adding a meal on top of it.

REFERENCE

Glossary

Ammonia	The first waste product of an animal's waste and decaying food, and highly toxic. The target reading is 0 ppm in a cycled tank.	Leucistic	The pale pink morph with dark eyes, the one most people picture. Pale skin, normal pigment in the eyes, and not the same as albino.
Baffle	Anything that breaks up a filter's outflow, from a piece of foam to a spray bar, so the current reaching the animal is gentle.	Morph	A color variety. Morphs differ in appearance only, never in care requirements.
Bsal	<i>Batrachochytrium salamandrivorans</i> , a chytrid fungus lethal to salamanders. The reason for the 2025 US import restrictions on genus <i>Ambystoma</i> .	Nitrate	The end product of the nitrogen cycle. Far less toxic than ammonia or nitrite, but keep it under 10 ppm with water changes.
Chiller	A device that actively cools aquarium water. The one cooling method that holds a target temperature regardless of the weather.	Nitrite	The intermediate product of the nitrogen cycle, also highly toxic. The target reading is 0 ppm.
Chloramine	A chlorine and ammonia compound used in many municipal supplies. Unlike chlorine it does not gas off if water is left standing, so a conditioner is required.	Paedomorphosis	Reaching adulthood while keeping larval features. An axolotl breeds at full size while still gilled and fully aquatic. Often called neoteny.
Cloaca	The single vent behind the back legs used for waste and reproduction. Its shape is the main way to sex an adult axolotl.	Quarantine	Housing a new animal separately, for several weeks, before it goes anywhere near an established one.
Cycling	Establishing the bacteria colony that converts ammonia to nitrite and nitrite to nitrate. Roughly four to six weeks in a new tank.	Regeneration	Regrowth of lost tissue. Axolotls regrow limbs, gill filaments, and parts of the heart, spinal cord and brain without scarring.
Dechlorinator	A water conditioner that neutralizes chlorine and chloramine in tap water before it reaches the animal.	Salt bath	A short therapeutic bath, 10 to 15 g of non-iodized salt per liter for 10 minutes, used against fungal infection alongside correcting the water.
Exophthalmia	Bulging of one or both eyes. In axolotls it is associated with prolonged exposure to high nitrate.	Saprolegnia	The fungal genus behind most of the white cottony growth seen on stressed or injured axolotls.
GH and KH	General hardness (dissolved minerals) and carbonate hardness (resistance to pH swings). Axolotls need mineral content and do badly in very soft water.	Sponge filter	An air-driven filter using a foam block. Gentle by design, which suits an animal that dislikes current, and it doubles as aeration in a power cut.
Impaction	A gut blockage from a swallowed object, most often gravel or oversized food. The preventable emergency of this species.	Tubbing	Keeping the animal in a plain container with a full daily water change instead of a filtered tank, used during cycling, quarantine, healing or treatment.

REFERENCE

Sources, version history & about

Where the numbers come from, so you can check them yourself.

Veterinary and research sources

- LafeberVet, Care of the Axolotl: size and lifespan, the preferred 60 to 64°F (15.6 to 17.8°C) range and the 75°F (24°C) illness threshold, pH 7.4 to 7.6, the ammonia, nitrite and nitrate targets, the 30 percent weekly change, filtration, substrate, cohabitation, feeding, and exophthalmia.
- Loh R, Common Disease Conditions in Axolotls, World Small Animal Veterinary Association proceedings, 2015: the salt bath dose of 10 to 15 g per liter for 10 minutes once or twice daily, the 41 to 59°F (5 to 15°C) treatment range, antibiotic classes and the caution against tetracyclines, and foreign body retrieval.
- VCA Animal Hospitals, Axolotls as Pets: the 75°F (24°C) threshold, portion timing, and bulging eyes.
- Ambystoma Genetic Stock Center, University of Kentucky, Guide to Axolotl Husbandry: the 60 to 65°F (15 to 18°C) colony range and the 72°F (22°C) ceiling, the pH band of 6.5 to 8, hard water, sexing, and maturity at about a year.
- IUCN Red List, *Ambystoma mexicanum*: Critically Endangered status and the Lake Xochimilco range. Page RB and others, BMC Genomics 9:78, 2008, for thyroid-induced metamorphosis.

Husbandry and regulatory references

- Aquarium Co-Op, on cooling a tank in a heat wave and during a power outage, for pages 9 and 35. Virginia Department of Wildlife Resources on the 2021 change, and US Fish and Wildlife Service, salamanders as injurious wildlife FAQ, for the 2025 Lacey Act listing, both on page 6. Beastly Facts axolotl guides: tank setup, feeding, health, handling, cost, enrichment, legal status and cold clean water.

Where sources disagree

Two figures are printed as ranges because credible sources differ. On temperature, veterinary guidance prefers 60 to 64°F (15.6 to 17.8°C) and a research colony targets 60 to 65°F (15 to 18°C), while the common hobby range runs to 68°F (20°C). This package uses 60 to 68°F with the cooler half preferred, since every source agrees on the ceiling. On adult size, vets give 9 to 10 in (23 to 25 cm) while larger individuals to about 12 in (30 cm) are documented, so both appear.

Version history

VERSION	TEMPLATE	WHAT CHANGED
2.0 · Sep 2026	t3	Rebuilt on the t3 template and expanded from 20 to 41 pages. Temperature, cooling, filtration, cycling, water chemistry and changes each got a page. New pages on legal status, choosing and sexing, cohabitation, arrival, growth, gill reading, tubbing and salt baths, power outages and a sitter sheet.
1.0 · Aug 2026	pre-t3	First edition, 20 pages.

This package is a husbandry reference compiled from the sources above and cross-checked against veterinary guidance at the time of publication. It is not a substitute for an amphibian or aquatic veterinarian, and nothing in it should delay a vet visit. Doses quoted here come from published veterinary references, so you can discuss them with a vet rather than treat an animal unsupervised.

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